

Literature Sources Analysis Using a Deep Impostor Approach

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Motivation: Why Internal Style Matters

- Most stylometric methods assume single-author, homogeneous documents.
- Real texts are often edited, compiled, or written over long periods.
- Internal stylistic variation is therefore ignored or treated as noise.
- Labeled data for stylistic boundaries is rarely available.

There is a need for an unsupervised system that detects internal stylistic changes.

Pipeline Execution

- Target text is segmented and embedded.
- Impostor-pair classifiers process each chunk.
- Outputs form continuous stylistic signals.
- Signal refinement emphasizes stylistic transitions.
- Consensus across impostors produces final boundaries.

Raw text is transformed into a 1D stylistic signal.

System Goal and Core Idea

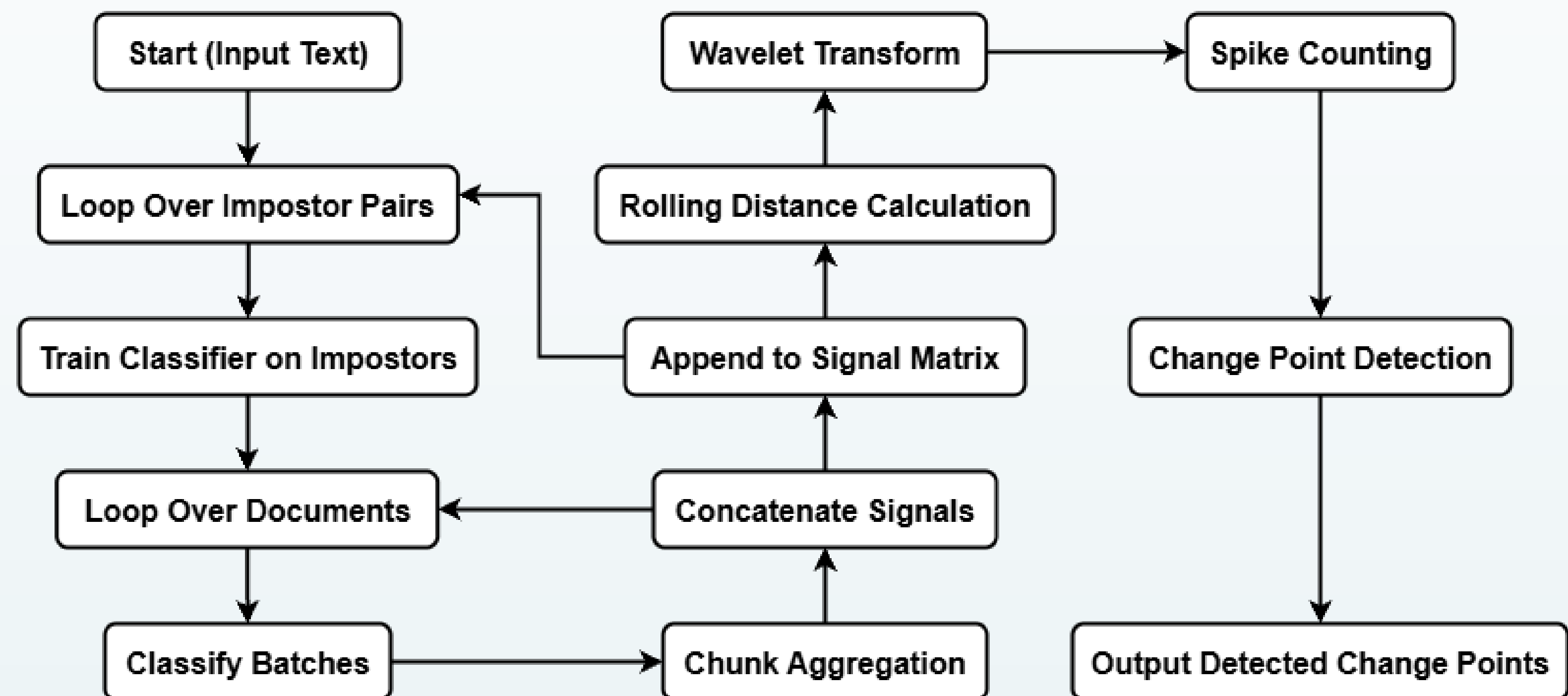
- Goal: Detect internal stylistic boundaries in long texts without supervision.
- No training is performed on the target document.
- Models are trained only on unrelated impostor texts.
- Classifier behavior is interpreted as a stylistic response.

Classifier behavior functions as a stylistic sensor rather than a predictor.

System Architecture

The system consists of four modular components:

- Text Segmentation**
Fixed-length batching and chunking of documents.
- Embedding Module**
Word2Vec or BERT representations.
- Stylistic Classification**
Deep binary classifiers trained on impostor pairs.
- Signal Analysis**
Signal construction and boundary detection



Validation: Proving the Method Works

- Validation performed before applying the method to Tanach.
- Shakespeare corpus provides expected stylistic variation.
- No training on Shakespeare texts.
- Clear stylistic transitions are detected.

Confirms detection of genuine stylistic change, not noise.

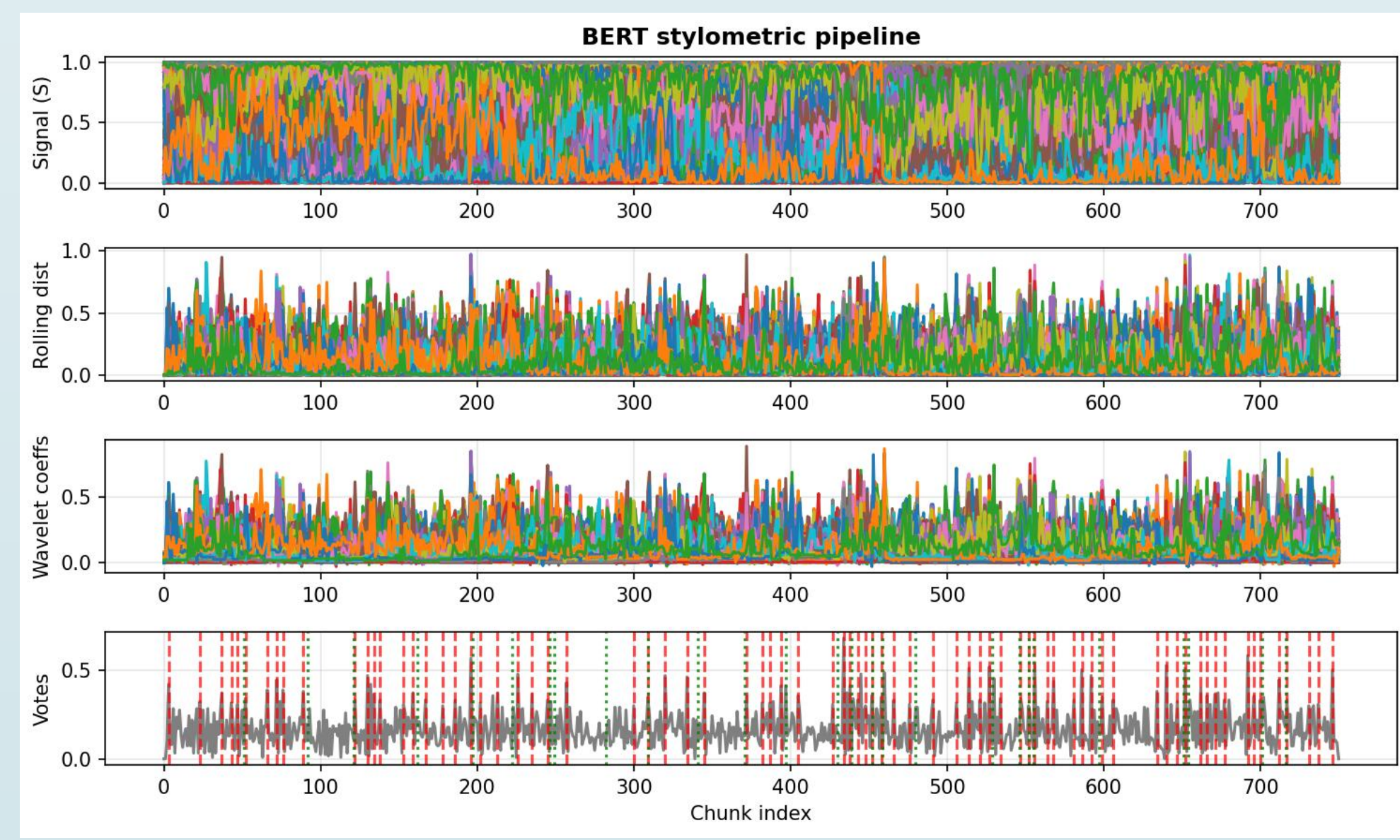
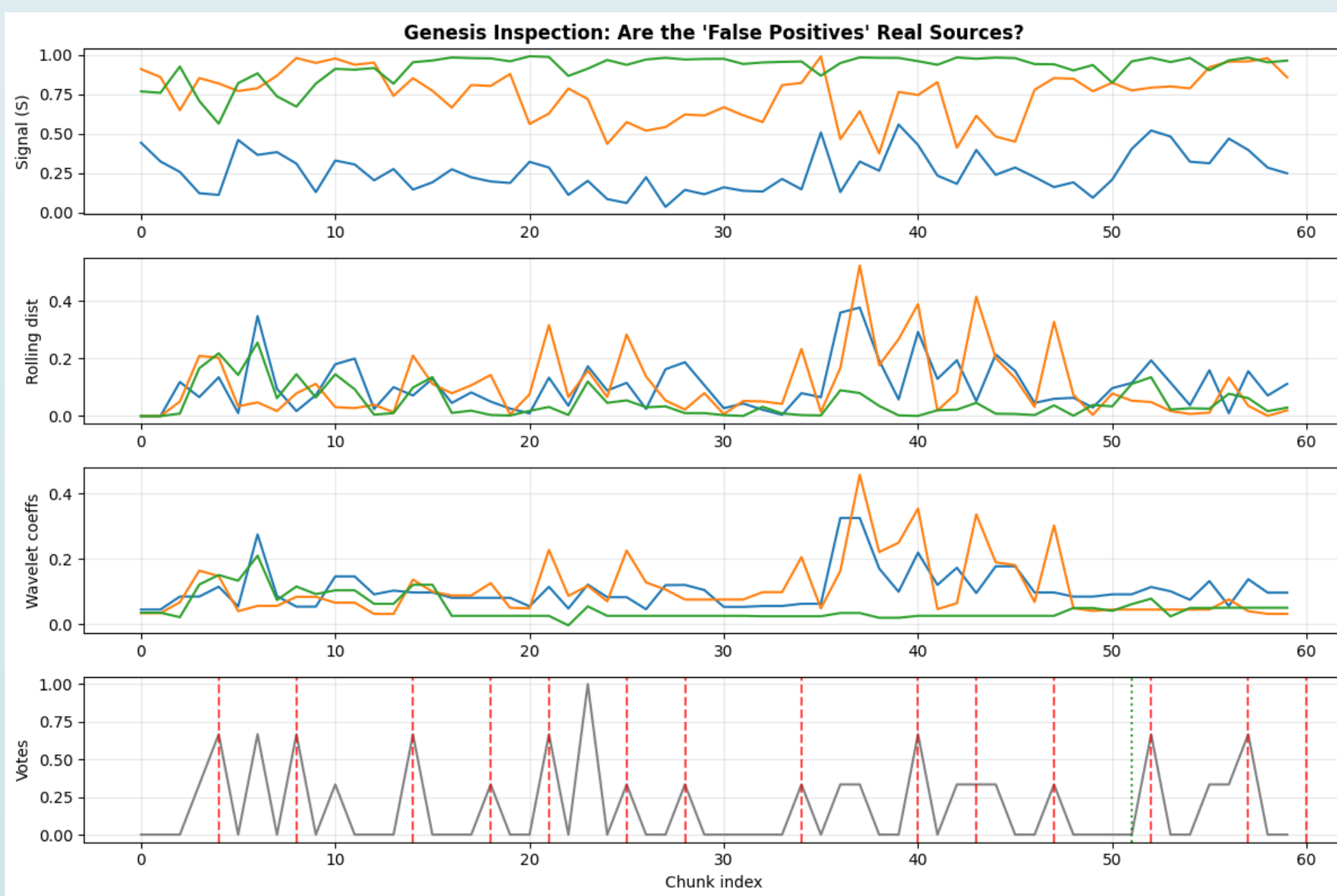
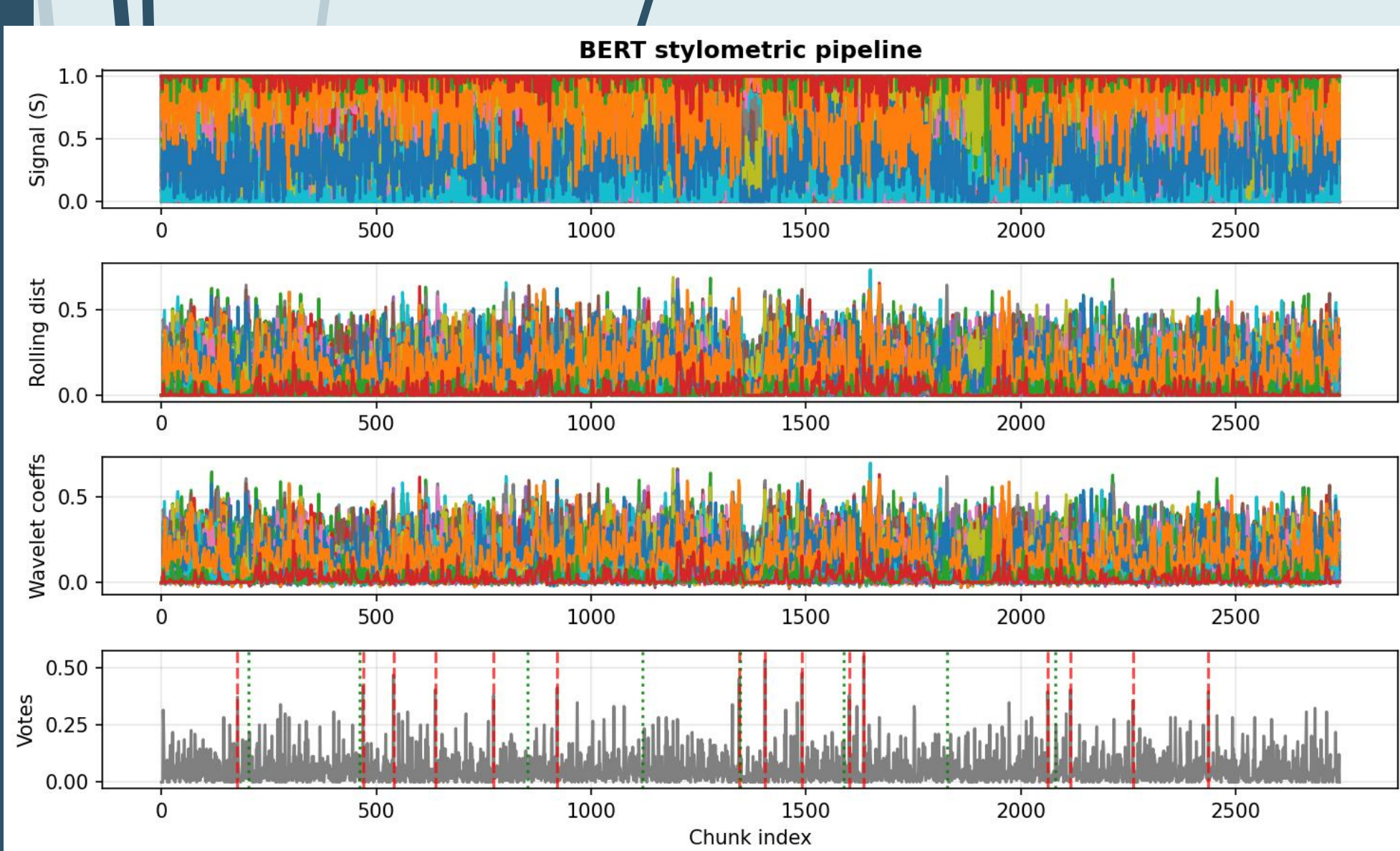
Conceptual Signal Analysis

- Each curve corresponds to a different impostor-pair classifier.
 - Signals differ in scale but reflect the same text region.
 - Flat regions indicate stylistic stability.
 - Aligned changes indicate potential boundaries.
- Stylistic structure emerges from agreement across signals.

Final Results

- The entire Tanach was analyzed as one continuous text.
- The system automatically identified points where writing style changes.
- Only a small number of clear boundaries were detected.
- These boundaries are consistent and easy to interpret.

The final output is an unsupervised stylistic segmentation of the Tanach.



Challenges and Constraints

- High computational cost due to multiple deep classifiers.
- Sensitivity to chunk size and segmentation choices.
- Tradeoff between noise suppression and boundary resolution.
- Design choices required balancing robustness and detail.



Book / Section	Detected Feature	Scholarly Significance (Analysis)
Genesis	Chunk 3 (Ch. 2-3)	Creation Sources: Shift from Priestly (Gen 1) to Yahwist (Gen 2) creation stories.
Isaiah	Chunk 586 (Ch. 40)	Deutero-Isaiah: Precise identification of the transition to "Second Isaiah".
Ezra / Daniel	Chunks 438, 700	Language Shift: Detection of Aramaic sections embedded within Hebrew text.
Psalms	Chunks 491-521	Internal Structure: Reflection of the five distinct books within Psalms.
Chronicles (1)	Chunk 382	Genre Shift: Transition from Genealogy lists (Ch. 1-9) to the Davidic narrative.
Samuel	Unified (FN)	Single Work: Validates the view of Samuel I-II as one continuous composition.

Conclusion

- The Deep Impostor model identified multiple internal stylistic transitions.
 - Without prior linguistic training, the system distinguished Hebrew and Aramaic passages.
 - Detected boundaries aligned with well-known scholarly observations, including:
 - stylistic divisions within Isaiah, separations between narrative and genealogical sections, and
 - differences between poetic and prose structures.
- These results show that unsupervised deep learning can serve as a data-driven tool for exploring textual structure in digital philology.

Can You Spot the Difference? (The Model Did)

Genesis 1
(The Priestly Source)
 בְּרֵאשִׁית בָּרָא אֱלֹהִים
 אֶת הַשָּׁמַיִם וְאֶת הָאָרֶץ...
 וַיֹּאמֶר אֱלֹהִים יְהי אוֹר
 אוֹר

The 'Smoking Gun':
 In Genesis 1, the text strictly uses 'Elohim' (God).
 In Genesis 2, it abruptly switches to 'YHWH Elohim' (Lord God).
 Our Deep Impostor model detected a massive signal spike exactly at this transition point (Chunk 3), proving it can identify distinct authors based on vocabulary and syntax patterns alone.

Genesis 2
(The Yahwist Source)
 בְּיוֹם עֲשׂוֹת יְהוָה
 אֱלֹהִים אָרָץ וְשָׁמַיִם...
 וַיִּצַּר יְהוָה אֱלֹהִים
 אֶת-הָאָדָם

(Style: Formal, Repetitive, uses 'Elohim')

(Style: Narrative, Personal, uses 'YHWH')